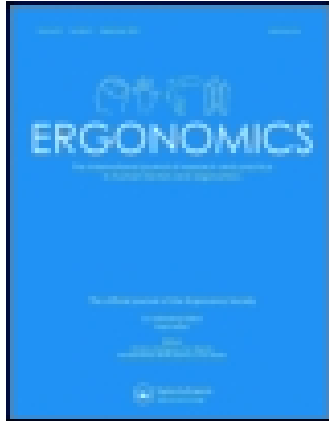




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TOM LEAMON

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Book Reviews

The Evolution of Health & Safety in Mines. By SIR ANDREW BRYAN (Garden City Press Ltd., 1975.) [Pp. 192] £5.90.

The field of interest covered is a fascinating one; a history with shocks and drama and yet recent enough to contribute to our understanding of attitudes in modern mining. It is a success story too; a steady improvement in health and safety is discernable over two centuries, culminating in the British Industry which is the safest in coal mining industries in the world today. Certainly in the 18th and 19th centuries this downward trend contains discontinuities which indicate significant changes every few years: 1812 (Felling Colliery Explosion 92 killed), 1844 (Hasswell Colliery 95 killed), Risca Explosion 28 killed and so on up to the present, when the Marcinelle Disaster in Belgium led directly to the establishment of the Mines Safety & Health Commission.

This book contains twelve chapters which cover the early developments in legislation, inquiries into accidents, employment of children, technical problems, developments in regulations, consolidation of law, the law and technical progress, recent enactments, Mines Inspectorate, health of miners, ancillary function and the Safety & Health of Persons at Work Act. The Reports, Acts, Inventions and organisational requirements for improving health and safety are outlined and often the initiating incident is identified.

However, in this book the craft of writing is not at all evident and it will require a keen student to persevere with this text and, I am afraid, such individuals will find the text lacking in detailed scholarship.

The book is author oriented and the readership does not appear to have been defined. Essentially the book is a record of the development of legislation (which has had a unique and recorded part to play in the improvement of work in coal mines), but the sections, such as "Ancillary Factors Helpful to Safety & Health", confuse rather than clarify the record of Enactments. Almost every page contains an abstract from a publication or even a couple of apparently verbatim comments from speeches in one of the Houses. Whilst this could be legitimate, an example illustrates the actual style which appears to offer no advantages:

"The Act empowered Inspectors of Mines appointed under it to enquire into, both above and below ground

'all Matters and Things connected with or relating to the Safety of the Persons employed'".

But as Boyd points in his book *Coal Mines Inspection*, the Act of 1850, although an important step in the right direction,

'did little more than provide for the appointment of inspectors who were few in number and not invested with much authority'."

The use of plates also appears indulgent; of 14 plates, two illustrate working conditions of children in the mid 19th century, two are facsimiles of Wage Bills 1758-1759 and no fewer than eight are portraits of researchers or legislators.

The most disturbing feature to me is the lack of emphasis on the role of the National Coal Board in reaching the present position. Health & Safety are accorded the greatest importance within the Industry and efforts of the Board, often hand in hand with the Inspectorate, deserves a better record than that provided by this book. I feel that this book is a disguised attempt to explain and justify the role of H.M. Chief Inspector of Mines—how much better it would have been if this had been done explicitly, for the past and present position of the Inspectorate are ones to be proud of, within and without the Industry.

TOM LEAMON.

Human Physiological Work Capacity. By R. J. SHEPHARD (London: Cambridge University Press, 1978.) [Pp. xiii + 303] £17.50.

In this, the 15th volume of a series reporting the work of the International Biological Programme, Roy Shephard sets out to draw together five years of research devoted to examining

factors influencing human work capacity. It should be said at once that he has succeeded in writing a lively book in which he has not allowed the mass of data to lull him into a 'review' style. Indeed, it is refreshing to have a book in which the personality and views of the author came through so forcibly, in which he does not hesitate to draw conclusions, to interpret and to reject from the impressive amount of data at his disposal.

If there are criticisms they are that there is undue concern with the Canadian (in particular the Eskimo and circumpolar) experience, reflecting both the author's own interests and work and the emphasis placed upon this region in the human adaptability programme. Nevertheless, it is a shame that greater effort was not made in the populous and developing world of Asia and Africa. Furthermore, if one of the central issues was "the degree to which individual differences in physical working capacity can be inherited", then the opportunity to obtain useful data was largely missed. Thus to Shephard's own question: "What attracted laboratory based exercise physiologists to exotic surroundings?" a partial answer is: seeing the world, as a bonus of research, adds spice to the experience. Indeed the lure is so strong that the obvious comparisons of performance of environmentally similar but genetically dissimilar populations now to be found in almost all big Western cities have virtually been ignored. Thus, Chapter 6, which concerns itself with work capacity and constitution is the slimmest in the book and refers to only two such major studies, Wyndham's among the Bantus and the Israeli studies of Davies and Glick.

The book is divided into 10 chapters. The first three: an introduction, sampling and methods, are of considerable general interest to all experimenters, particularly those setting out on the difficult task of human experimentation in the field. The paramount importance of sampling without bias or the awareness of what bias is being introduced is well made. The difficulties of coming to any agreement on the standardisation of methods among tetchy and even sometimes egocentric academics is underlined. Is it idle to muse how much more might be achieved in applied research if there was greater standardisation of methods while still allowing, with some curbing, the natural exuberance to seek novelty?

The next three chapters are a discussion of the various factors that may or may not influence physical working capacity such as climate, geography, socio-economic status or genes. Some of the arguments are very sophisticated and specialised, while others such as those concerning the assessment and consequences of malnutrition should have wide appeal. The length of Chapter 7 on the performance of athletes reflects that almost obsessive interest of physiologists to measure extremes in the belief that it enables them to delineate "the full possible range of physical fitness in various societies". It is, however, clear that many so-called primitive or even developing societies have not yet joined in the bizarre battle for supremacy on the track or in the baths, thus their full potential has not yet been realised.

Chapters 8 and 9 are of outstanding interest, describing the growth and decline of working capacity as man matures and ages.

The Epilogue (in Chapter 10) is made up of a series of apologia for the inadequacies of the human adaptability programme. While there are inevitable shortcomings, there is much to be proud of. In fact the Epilogue smacks of a plea for the continuation of the species *homo physiologia humanis* so that their endeavours may continue and their visits to exotic places may continue to be financed. There is a very low key advocacy of the implications and application of all the acquired knowledge. It would have been in line with the bulk of the book if Professor Shephard had given us his views on whether it was all worthwhile and whether we now knew more about the overall theme 'the biological basis of productivity and human welfare'. It is no longer sufficient to gather data, we must indicate where it is leading us and how it could be applied. It is not enough to jump on the ecological bandwagon and rest content that the data so arduously collected is just another warning "of the ultimate dangers associated with life in an over-sheltered habitat".

What then of the work as a whole and to whom is it relevant? It is a masterly summary, a textbook for a long time to come, dealing with work capacity. Your reviewer stands back in awe, to admire an author, who quotes more than 700 published references, yet makes much of his book lively, controversial and above all readable. All students of the human sciences (social as well as medical) could benefit from Chapters 1-3. They will gain further insight into the intricacies of human work capacity by judiciously seeking out from the remainder areas of more generalized interest. For applied physiologists, students and teachers, the book will surely become a *sine qua non*.

RAINER GOLDSMITH.